

Visible-light-active eco-friendly MgO-CeO₂ nanocomposites for efficient degradation of carcinogenic dyes

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ABSTRACT

In this study, Magnesium oxide (MgO) and cerium oxide (CeO₂) nanoparticles (NPs) were synthesized independently using a green route, employing aqueous *Saraca asoca* leaf extract as a reducing and stabilizing agent. MgO-CeO₂ nanocomposites (NCs) were subsequently fabricated via physical grinding and calcined at different temperatures to improve crystallinity and tailor optical properties. The structural, morphological, and physicochemical characteristics of the as-synthesized NCs were systematically investigated using XRD, UV-DRS, HR-XPS, FT-IR, FE-SEM, EDX, HR-TEM, TGA, BET, and zeta potential analyses. XRD and TGA results confirmed the formation of a mesoporous heterojunction with a face-centered cubic (fcc) structure, while HR-TEM revealed mixed morphologies with an average particle size of ~20.7 nm. Optical studies demonstrated enhanced visible-light absorption with a reduced band gap of 2.65 eV. Under 500 W visible-light irradiation, the MgO-CeO₂ NCs exhibited excellent photocatalytic activity, achieving 94.7% degradation of Rhodamine B and 91.0% degradation of Eosin Y, following first-order kinetics. Radical scavenger experiments identified superoxide (•O₂⁻) and hydroxyl (•OH) radicals as the dominant reactive species, and LC-MS analysis confirmed stepwise mineralization into low-molecular-weight intermediates. The NCs also showed good recyclability and stability over multiple cycles. Future studies should focus on evaluating catalyst performance in real wastewater matrices under continuous-flow conditions, improving long-term durability, and enabling scalable or immobilized catalyst systems. Extending the application toward complex pollutant mixtures, including pharmaceuticals, pesticides, and heavy metals, will further strengthen their potential for practical environmental remediation.

1. Introduction

Nanocomposites (NCs) have gained significant interest because of their distinct characteristics, which often differ significantly from their bulk counterparts. Typically, smaller than 100 nm, nanomaterials exhibit enhanced surface area, porosity, and size-dependent properties, opening new avenues in materials science, chemistry, biology, and device fabrication [1–4]. Among these, metal oxide nanoparticles (NPs) are particularly promising as photocatalysts because of their strong redox potential, high reactivity, stability under harsh conditions, and tunable electronic properties. A variety of nanomaterials like graphitic carbon nitride, zinc oxides, copper oxides, cobalt oxides, manganese oxides, iron oxides, tungsten oxides, tin oxides, cadmium sulphides, copper sulphides, tin sulphides, zirconium sulphides, and other

compounds have been extensively explored for their photocatalytic degradation capabilities [5–8]. However, rapid recombination of photo-excited charge carriers in many oxide semiconductors hinders their photocatalytic activity, motivating the development of mixed oxide NCs [9,10].

Magnesium oxide (MgO) is a non-toxic, widely available, and environmentally safe broad-bandgap oxide known for its excellent physical, chemical, and optoelectronic properties [11]. It features a high surface area, moderate dielectric constant, low refractive index, strong surface reactivity, large adsorption capacity, and is easy and cost-effective to synthesize [9,12,13]. However, its photocatalytic effectiveness under visible light is restricted due to a broad bandgap as well as rapid recombination of charge carriers produced by photons [14,15]. It has applications in catalysis, paints, superconductors, and wastewater

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treatment. Nevertheless, its photocatalytic performance is constrained by limited electron-hole separation. The development of MgO composites has received considerable attention in recent years. Furthermore, the electron-hole separation performance of MgO-based photocatalysts must be improved. In general, creating a new mixed oxide is an effective strategy for increasing the host material's surface area and reducing its bandgap, therefore strengthening the material's ability to absorb visible light along with better photocatalytic activity [13].

Cerium dioxide (CeO_2), a rare-earth oxide, exhibits high chemical stability, oxygen-ion mobility, and reversible $\text{Ce}^{4+}/\text{Ce}^{3+}$ redox transitions, which enhance its catalytic potential. Quasi-free localized electrons and oxygen vacancies in CeO_2 induce mixed ionic-electronic conductivity. CeO_2 has been widely investigated for heterogeneous catalysis, solid oxide fuel cells, environmental remediation, as well as lithium-ion batteries [16–18]. CeO_2 NPs with high crystallinity have been synthesized using conventional processes like sol-gel, chemical precipitation, and hydrothermal techniques. However, these techniques often require expensive equipment, long processing times, high energy input, and hazardous chemicals [19]. The excellent photocatalytic activity of CeO_2 is attributed to its outstanding optical and physicochemical characteristics, low production cost, and superior catalytic performance [20]. Nevertheless, its practical application under visible light is limited owing to quick electron-hole recombination and a wide energy gap that reduces its ability to capture visible light [21].

MgO was specifically selected as the coupling partner for CeO_2 due to its non-toxic nature, high surface area, and strong surface reactivity, which allow it to form stable heterojunctions. Previous studies on visible-light photocatalysts have shown that transition-metal oxides such as NiO and CuO can enhance photocatalytic activity through improved charge separation in heterojunction systems; however, these materials often involve more complex architectures and potential stability concerns [22]. In contrast, MgO, an alkaline-earth oxide, effectively promotes interfacial charge-carrier separation while maintaining chemical stability, low cost, and simple synthesis [11,16]. The formation of the MgO- CeO_2 heterojunction facilitates efficient charge separation, while Ce^{3+} ions, arising from the partially filled and shielded 4f electronic states of cerium, together with oxygen-vacancy-related defect states at the interface, introduce intermediate energy levels, extending light absorption into the visible region [13]. Consequently, the enhanced visible-light photocatalytic activity of MgO- CeO_2 NCs arises primarily from interfacial charge transfer and defect-assisted electronic effects, rather than from the intrinsic bandgap of MgO alone.

Synthetic dyes like Rhodamine B (RhB) and Eosin Y (EY) are major contributors to water contamination. RhB, a member of the xanthene dye group, finds broad application in the textile industry and biological studies. It displays vivid color, high persistence in diluted solutions, and produces hazardous intermediates during degradation. Under limited oxygen conditions, it transforms into aromatic amines that are toxic and cancer-inducing, endangering ecosystems and human health. In aquatic systems, RhB disrupts microbial balance, hinders photosynthesis in marine plants, reduces dissolved oxygen levels, and adversely affects aquatic life. Short-term exposure may irritate the eyes, throat, and respiratory tract, while prolonged exposure can lead to reproductive as well as growth disorders in both human beings and animals. Owing to its stable nature and difficulty in decomposition, RhB is extensively used as a reference toxin in studies on photocatalytic decomposition [19].

Similarly, EY, another synthetic dye used in textiles and formerly in food colouring, poses environmental and health hazards, including eye irritation, organ toxicity, and potential carcinogenicity. Although banned for food use in several countries, such as the United States, it remains in circulation in many developing regions with less stringent regulatory control [23,24]. The unregulated discharge of these dyes into aquatic environments presents a significant ecological and public health challenge. Hence, this study focuses on developing efficient strategies for detecting and removing such toxic dyes from polluted water systems [25]. Advanced oxidation processes, particularly photocatalysis, are

effective strategies for degrading these persistent pollutants [26].

Green synthesis using plant extracts provides an environmentally friendly and sustainable route to metal oxide NPs with high surface area, good crystallinity, and enhanced photocatalytic activity [11,27]. Compared to conventional hydrothermal or sol-gel methods, which often require large amounts of hazardous organic solvents and higher energy input, this approach is simpler, less chemically intensive, and more environmentally sustainable. Although *Saraca asoca* (Sa) leaf-mediated synthesis of individual MgO and CeO_2 NPs and their UV-driven photocatalytic activities have been previously reported [11,19], the present work advances beyond earlier studies by integrating the green-synthesized oxides into MgO- CeO_2 NCs via a simple physical grinding route under ambient conditions. The Sa leaf extract, rich in polyphenols, flavonoids, alkaloids, terpenoids, steroids, glycosides, and carbohydrates, acts as a natural reducing and stabilizing agent during nanoparticle formation [11,28]. The formation of the MgO- CeO_2 heterojunction enables effective bandgap engineering, narrowing the optical gap from 4.08 eV (MgO) and 2.78 eV (CeO_2) to 2.67 eV, together with enhanced surface characteristics and improved interfacial charge separation, thereby extending light absorption into the visible region [13]. Unlike earlier MgO- or CeO_2 -based photocatalysts that mainly emphasize single-dye degradation, this study demonstrates efficient degradation of both cationic (RhB) and anionic (EY) dyes under UV and visible light irradiation [25,26]. In addition, the photocatalytic mechanism is validated through reactive species trapping experiments and LC-MS confirmed mineralization, confirming true degradation rather than mere chromophore bleaching. Quantitatively, the MgO- CeO_2 NCs exhibit markedly higher degradation efficiencies and rate constants than pristine MgO and CeO_2 under both UV and visible light irradiation, highlighting the synergistic role of heterojunction-driven charge transfer and defect-assisted electronic states.

2. Experimental section

2.1. Raw materials

All reagents employed in this research were of analytical grade and required no further purification. Magnesium nitrate hexahydrate ($\text{Mg}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$) was obtained from Sisco Research Laboratory, Maharashtra, India, while 99% cerium nitrate hexahydrate ($\text{Ce}(\text{NO}_3)_3 \cdot 6\text{H}_2\text{O}$) was procured from Spectrochem Pvt. Ltd., Mumbai. Sodium hydroxide flakes (NaOH , 97%) were supplied by Molychem, Mumbai, and 99.9% ethanol ($\text{CH}_3\text{CH}_2\text{OH}$) was obtained from Zenith Biochemical Industries, Pune, Maharashtra. Rhodamine B ($\text{C}_{28}\text{H}_{31}\text{ClN}_2\text{O}_3$) and Eosin Y ($\text{C}_{20}\text{H}_6\text{Br}_4\text{Na}_2\text{O}_5$) dyes were purchased from Avra Synthesis, Hyderabad, Telangana. Double-distilled water (DDW) served as the solvent for all aqueous solutions. Fresh Sa leaves were collected around Thrissur, Kerala, and their identification was verified by the Botanical Survey of India, Coimbatore, Tamil Nadu. This plant was selected due to its rich composition of phytochemicals and its suitability as a natural reducing and stabilizing agent for the synthesis of NPs.

2.2. Synthesis of aqueous *Saraca asoca* leaf extract

Fresh Sa leaves were carefully rinsed with distilled water to remove surface contaminants and then shade-dried for 10–15 days to preserve their phytochemical content. The dried leaves were finely ground to obtain a uniform powder. Approximately 5 g of this powdered material was mixed with 100 mL of DDW to prepare the extract. The mixture was gently boiled at 80 °C for 5–6 h until its volume was reduced by approximately half. To remove any remaining plant debris, the solution was filtered twice after cooling to ambient temperature. The clarified extract was stored at 4 °C and acted as both a natural reducing and stabilizing agent during the nanoparticle synthesis process [11,19]. Qualitative phytochemical screening of the aqueous Sa leaf extract was performed following standard procedures reported in the literature

[27]. The presence of major phytochemical groups, including polyphenols/tannins, flavonoids, alkaloids, terpenoids, glycosides, steroids, and carbohydrates, was confirmed.

2.3. Synthesis of MgO NPs

MgO NPs were prepared through a green method using *Sa* leaf extract, which acted as both a reducing and capping agent. An aqueous mixture of 0.1 M $\text{Mg}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ and the prepared leaf extract was maintained at 80 °C for 2 h, and alkaline conditions were generated by the gradual addition of 0.1 M NaOH solution. NaOH was used solely to adjust the reaction medium to basic conditions, facilitating oxide formation. The appearance of a green precipitate indicated the successful formation of MgO NPs. To remove any remaining plant debris, the precipitate was centrifuged and thoroughly washed with DDW. The product was dried in a hot-air oven at 60 °C and then calcined at various temperatures (200, 300, 500, 700, and 800 °C) to remove residual organics, enhance crystallinity, and tune the bandgap. During calcination, the dried MgO NPs gradually changed color from green to colorless [11].

2.4. Preparation of CeO_2 NPs

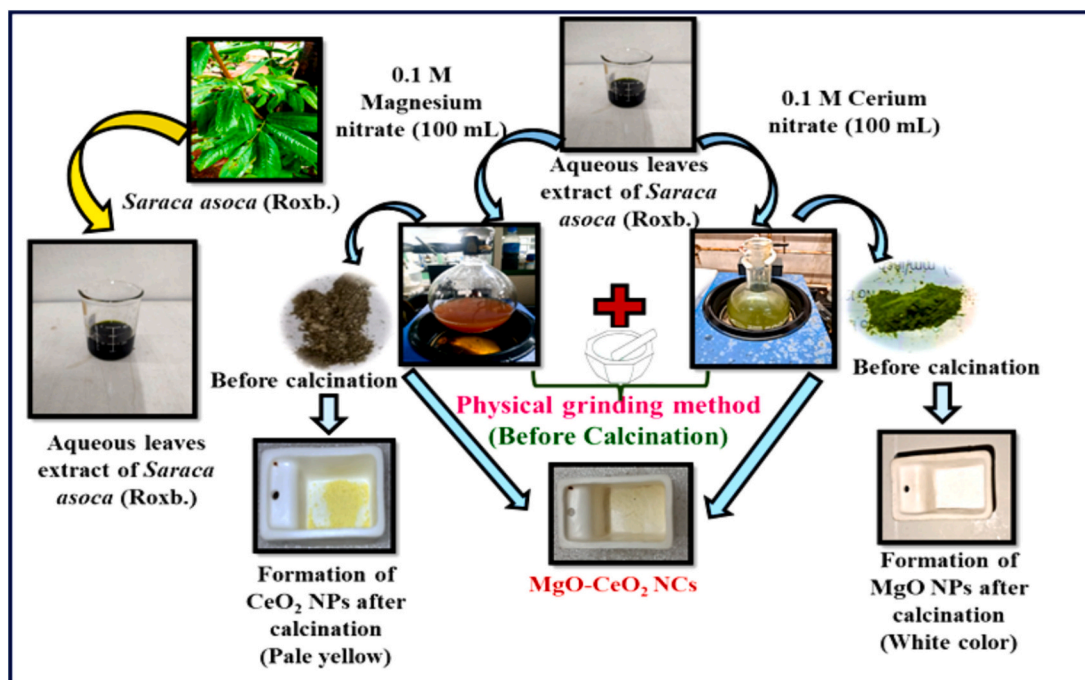
CeO_2 NPs were synthesized using a green approach, with *Sa* leaf extract serving as both a reducing and stabilizing agent. A 0.1 M solution of $\text{Ce}(\text{NO}_3)_3 \cdot 6\text{H}_2\text{O}$ (4.34 g) was prepared in 100 mL of DDW. To this solution, 10 mL of *Sa* extract was added in a 1:10 ratio with the cerium salt. The mixture was stirred and boiled at 80 °C for 2 h. The pH of the solution was gradually adjusted to 12 using 0.1 M NaOH, which was employed solely to provide basic conditions necessary for oxide formation. This adjustment resulted in a visible color change from colorless to brown, indicating the formation of cerium hydroxide ($\text{Ce}(\text{OH})_3$). The precipitate was recovered by centrifugation and washed several times with distilled water and ethanol to remove any residual reactants. The purified material was then calcined at various temperatures (300–800 °C), producing pale yellow CeO_2 NPs with enhanced crystallinity and optical properties [19].

2.5. Fabrication of MgO-CeO_2 NCs

MgO-CeO_2 NCs were prepared via a simple physical grinding approach, as illustrated in Scheme 1. Pre-synthesized MgO and CeO_2 NPs were manually mixed in varying weight ratios using a grinding apparatus to achieve a homogeneous dispersion. The mixtures were then thermally treated at 500, 600, 700, and 800 °C to remove residual organic compounds, enhance crystallinity, and promote phase formation, which also allowed tuning of the optical bandgap [25]. The structural, morphological, and optical properties of the NCs were subsequently characterized using multiple analytical techniques. Finally, the photocatalytic performance of the MgO-CeO_2 NCs was evaluated for the degradation of RhB (cationic) and EY (anionic) dyes under visible light, demonstrating their effectiveness in dye removal.

2.6. Photocatalytic study of MgO-CeO_2 NCs

The photocatalytic activity of MgO-CeO_2 NCs under visible light was investigated for the degradation of two model organic dyes: cationic RhB and anionic EY. Experiments were conducted to examine the effects of key parameters, including solution pH, dye concentration, catalyst dosage, and light intensity. In a typical experiment, 50 mL of 10 ppm RhB or EY solution at neutral pH (7) was treated with 1 mg/mL of MgO-CeO_2 NCs. The dye-catalyst suspensions were ultrasonically dispersed for 10 min and then kept in the dark (RhB: 1 h; EY: 30 min) to establish adsorption-desorption equilibrium. The negligible dye loss observed during this dark period confirmed that physical adsorption contributed minimally to overall degradation. Photocatalytic degradation was subsequently carried out in an annular quartz photoreactor illuminated with a 500 W, 265 lx visible-light source (400–800 nm), with continuous aeration provided by a mini air pump to maintain suspension and oxygenation. Aliquots of 2 mL were withdrawn at predetermined intervals, centrifuged to remove catalyst particles, and the supernatants were analyzed using a JASCO V-670 UV-vis spectrophotometer. Dye concentrations were monitored at $\lambda_{\text{max}} = 554$ nm for RhB and $\lambda_{\text{max}} = 515$ nm for EY. The temporal decrease in dye concentration was used to construct degradation profiles and calculate the apparent rate constant, assuming first-order kinetics (Eq. 1),



Scheme 1. Preparation of MgO-CeO_2 NCs using a simple physical grinding method.

$$\ln(C/C_0) = -kt \quad (1)$$

Here, C_0 is the initial dye concentration, C is the concentration at any time t , and k is the rate constant.

The photocatalytic activity of MgO-CeO₂ NCs was assessed according to (Eq. 2),

$$\text{Photocatalytic efficacy (\%D)} = [(A_0 - A_t)/A_0] \times 100 \quad (2)$$

Here, A_0 and A_t represent the absorbance of RhB or EY at the beginning and at a given time during the reaction, respectively.

2.7. Quenching study of photocatalytic reactive species

Scavenger studies were performed to identify the primary reactive species responsible for the photocatalytic degradation of RhB and EY using MgO-CeO₂ NCs. Specific chemical scavengers were added individually to the reaction system to selectively quench targeted reactive species generated under visible-light irradiation. Sodium nitrate (NaNO₃) served as an electron (e^-) scavenger, while sodium azide (NaN₃) quenched singlet oxygen (1O_2). Terephthalic acid (TPA) was used for detection of hydroxyl radicals ($\bullet OH$), and p-benzoquinone (PBQ) acted as a scavenger for superoxide radicals ($\bullet O_2^-$). Ethylenediaminetetraacetic acid (EDTA) was employed to capture photoinduced holes (h^+) and evaluate their contribution.

All scavenger experiments were conducted under identical conditions, maintaining equal amounts of each scavenger and MgO-CeO₂ NCs to allow reliable comparison. The catalyst loading was fixed at 1 mg/mL, and 1 mg of each scavenger (NaNO₃, NaN₃, TPA, PBQ, or EDTA) was added separately to the reaction system, while keeping the dye solution volume and light irradiation constant. Following scavenger addition, the photocatalytic performance of the MgO-CeO₂ NCs was monitored by measuring the degradation efficiency of RhB and EY under the same experimental conditions. A significant decrease in dye degradation in the presence of a particular scavenger indicated the major involvement of the corresponding reactive species. This systematic study provided insight into the primary reactive species responsible for the visible-light-driven photocatalytic activity of MgO-CeO₂ NCs.

2.8. Material characterization of MgO-CeO₂ NCs

The physicochemical and microstructural properties of the green-synthesized MgO-CeO₂ NCs were comprehensively characterized using a suite of analytical techniques. UV-visible diffuse reflectance spectra (UV-DRS) were recorded using a JASCO V-670 spectrophotometer to determine the optical bandgap. Functional groups were identified via Fourier-transform infrared (FT-IR) spectroscopy (Shimadzu IR-Affinity-1, 400–4500 cm⁻¹). X-ray diffraction (XRD, Bruker D8 Advance, Cu K α , 30 kV, 40 mA) was employed to examine the crystalline structure. Thermal stability was evaluated by thermogravimetric analysis (TGA, TA Instruments SDT Q600) from ambient temperature to 1200 K at a heating rate of 20 °C/min under a nitrogen atmosphere. Surface charge was assessed using zeta potential measurements (Horiba SZ-100), while morphology and elemental composition were analyzed via field-emission scanning electron microscopy (FE-SEM, Thermo Fisher Quanta 250 FEG) coupled with energy-dispersive X-ray spectroscopy (EDX, Oxford Instruments). High-resolution transmission electron microscopy (HR-TEM, FEI Tecnai G2 F30 S-TWIN, 300 kV) provided detailed information on particle size and morphology. Specific surface area and porosity were determined using the BET method (Autosorb-iQ). Surface elemental states were probed using high-resolution X-ray photoelectron spectroscopy (HR-XPS, PHI VersaProbe II, Al K α source). Photoluminescence spectroscopy (PL) was used to detect optical and electronic properties (HITACHI, F-7000). Photocatalytic experiments were conducted in an annular quartz photoreactor under different irradiation sources, and the light intensity at the reactor surface was measured for each condition. UV irradiation was provided by a 365 nm

mercury lamp (250 W) with an intensity of 380 lx, while visible-light irradiation (400–800 nm) was supplied by tungsten lamps of 500 W and 300 W, corresponding to intensities of 265 lx and 59 lx, respectively. Natural sunlight was also employed as an irradiation source. The degradation of dyes was monitored using UV-visible spectrophotometry, and liquid chromatography-mass spectrometry (LC-MS, Waters Xevo G2-XS QToF) was used to identify intermediate degradation products.

3. Results and discussions

3.1. XRD analysis

The crystalline structure and phase composition of the prepared MgO-CeO₂ NCs were examined using XRD. Figs. S2a and S2b present the XRD patterns of MgO-CeO₂ NCs prepared with varying weight ratios, while Fig. 1a shows the effect of calcination temperature on the crystallization of the optimized 1:1 MgO-CeO₂ NCs. For CeO₂, well-defined diffraction peaks were observed at 2θ values of approximately 28.55°, 33.07°, 47.49°, 56.32°, 59.05°, 69.42°, 76.74°, 79.02°, and 88.45°, corresponding to the (111), (200), (220), (311), (222), (400), (331), (420), and (422) lattice planes. These reflections are characteristic of a fluorite-type face-centered cubic (fcc) structure and match JCPDS card no. 00-004-0593, which reports a lattice parameter of 5.411 Å [19].

For MgO, prominent peaks appeared at 36.94°, 42.90°, 62.30°, 74.67°, and 78.61°, assigned to the (111), (200), (220), (311), and (222) planes of the fcc structure, in agreement with JCPDS 00-004-0829, which reports a lattice parameter of 4.213 Å [11,25]. The presence of sharp, well-defined peaks in the MgO-CeO₂ NCs confirms their high crystallinity. In Fig. 1a, MgO peaks are marked with (♥), while CeO₂ peaks are marked with (●) for clear phase identification. XRD analysis indicated that with increasing CeO₂ content in the MgO matrix, the intensity of MgO related peaks gradually decreased, while CeO₂ peaks became more prominent. Slight peak shifts were also observed, which can be attributed to variations in temperature, lattice stress or strain, MgO incorporation into the CeO₂ lattice, and differences in crystallite size [29,30]. No additional or unexpected peaks were detected, confirming that the synthesized MgO-CeO₂ NCs are phase-pure, consisting of two distinct crystalline phases rather than a solid solution. The average crystallite size of the NCs was estimated using the Scherrer equation (Eq. 3) [11,25].

$$D = k\lambda/\beta\cos\Theta \quad (3)$$

Here, β represents the full width at half maximum (FWHM) of the (hkl) diffraction peaks in radians, λ (1.5406 Å) is the wavelength of Cu K α radiation, k is the Scherrer constant (typically 0.9), and Θ is the Bragg angle. MgO-CeO₂ NCs calcined at 500 °C exhibited an average crystallite size of 17.96 nm, with a corresponding bandgap of 2.87 eV. Increasing the calcination temperature to 600 °C led to a slightly larger crystallite size of 18.02 nm and a decrease in bandgap to 2.77 eV. At 700 °C, the crystallite size increased more noticeably to 20.32 nm, accompanied by a further narrowing of the bandgap to 2.67 eV. At 800 °C, the crystallite size increased modestly to 20.55 nm, while the bandgap showed a slight widening to 2.80 eV. These results indicate that higher calcination temperatures enhance the crystallinity of the NCs, as reflected by sharper and more intense diffraction peaks [11,19]. Table 1 summarizes the crystallite sizes and bandgap energies of MgO-CeO₂ NCs prepared with different weight ratios and calcination temperatures.

3.2. UV-vis spectroscopy analysis

A key factor in evaluating the photocatalytic performance of a synthesized material is its ability to absorb light over a broad spectral range. The optical properties of MgO-CeO₂ NCs were investigated using UV-vis DRS in the 200–800 nm range (Fig. 1). The *Sa* leaf extract exhibits characteristic absorption bands at 231 nm and 276 nm (Fig. 1b),

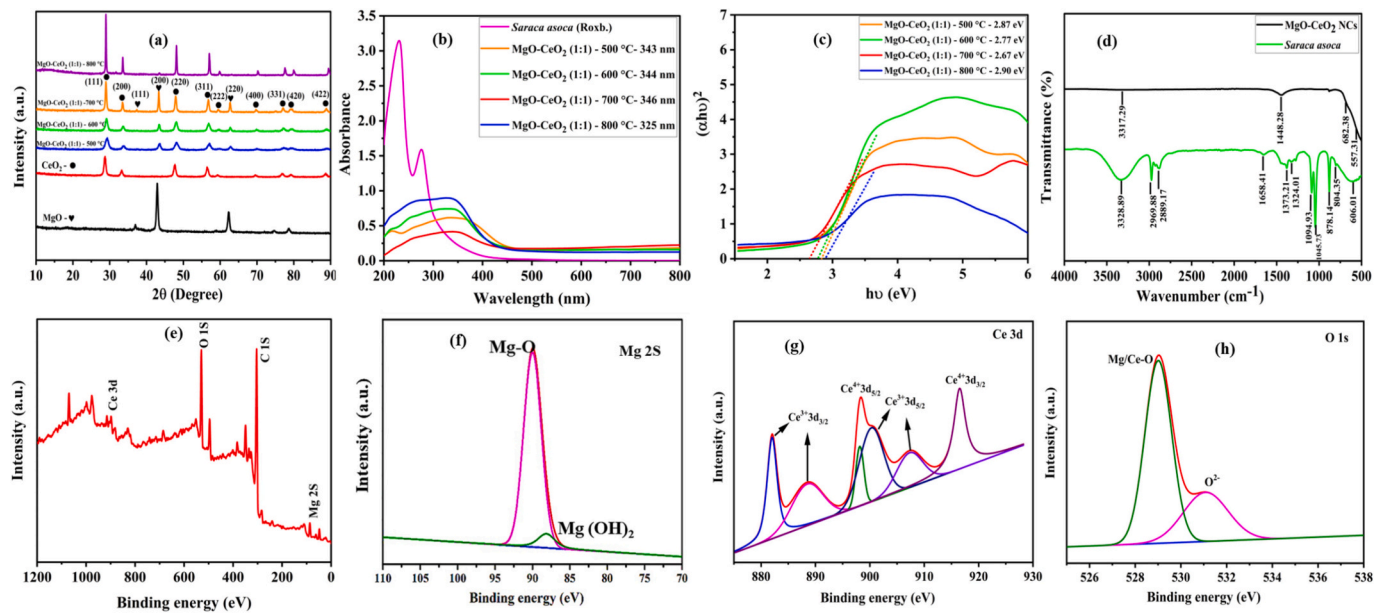


Fig. 1. (a) XRD pattern of MgO-CeO₂ NCs (1:1), (b) UV-vis-DRS spectra of *Sa*-leaf extract and MgO-CeO₂ NCs, (c) Tauc plot of MgO-CeO₂ NCs (1:1) at various calcination temperatures, (d) FT-IR spectra of MgO-CeO₂ NCs (1:1), (e) HR-XPS of the wide spectrum of MgO-CeO₂ NCs (1:1), (f) HR-XPS of Mg 2s, (g) HR-XPS of Ce 3d and (h) HR-XPS of O 1s.

Table 1

The crystallite size with bandgap measurements for MgO-CeO₂ NCs.

MgO-CeO ₂ NCs	Crystallite size in nm	Bandgap in eV
Various weight proportions		5.60
MgO-CeO ₂ (9:1)	7.19	3.13
MgO-CeO ₂ (4:1)	7.32	3.04
MgO-CeO ₂ (7:3)	7.47	2.98
MgO-CeO ₂ (3:2)	8.29	2.93
MgO-CeO ₂ (1:1)	17.96	2.87
MgO-CeO ₂ (2:3)	11.05	2.95
MgO-CeO ₂ (3:7)	8.60	3.00
MgO-CeO ₂ (1:4)	7.03	3.04
MgO-CeO ₂ (1:9)	5.60	3.09
Various calcination temperatures (°C)		
MgO-CeO ₂ (1:1) - 500 °C	17.96	2.87
MgO-CeO ₂ (1:1) - 600 °C	18.02	2.77
MgO-CeO ₂ (1:1) - 700 °C	20.32	2.67
MgO-CeO ₂ (1:1) - 800 °C	20.55	2.80

corresponding to polyphenolic compounds that mediate metal ion reduction and stabilize NPs during synthesis [11,19]. Figs. S3a-d present the UV-vis DRS spectra and the corresponding Tauc plots for MgO-CeO₂ NCs with varying weight ratios. Pure MgO and CeO₂ NPs showed bandgaps of 4.08 eV [11] and 2.78 eV [19], respectively. The MgO bandgap obtained in this work (4.08 eV) is comparable to previously reported values for chitosan-assisted MgO systems (4.15–4.30 eV), suggesting a similar contribution of defect-related localized states induced during synthesis [31]. Upon coupling MgO with CeO₂ in a 1:1 ratio, the bandgap decreased significantly to 2.87 eV, confirming the formation of MgO-CeO₂ heterostructures. This red shift reflects strong interfacial interactions between MgO and CeO₂. The enhanced absorption across the visible region is likely due to the synergistic combination of the two oxides, which extends light harvesting, suppresses electron-hole recombination, and improves photocatalytic efficiency [26]. The effect of calcination temperature on the optical properties of the composites was also investigated (Fig. 1b). Samples with a 1:1 ratio were heat-treated at 500, 600, 700, and 800 °C to tune the bandgap (Fig. 1c). The bandgap narrowed from 2.87 eV to 2.67 eV as the absorption edge shifted toward longer wavelengths between 500 and 700 °C. Further

heating to 800 °C caused a slight blue shift, increasing the bandgap to 2.90 eV. Bandgap values for both the composites and individual oxides were derived from Tauc plots (Fig. S4) using the Tauc relation (Eq. 4) [25]. The observed bandgap narrowing with increased CeO₂ content is attributed to the inherently lower bandgap of CeO₂ and ionic diffusion at grain boundaries, which promotes bandgap reduction and particle growth [1].

$$(\alpha h\nu)^n = A(h\nu - E_g) \quad (4)$$

Here, $h\nu$ is the photon energy, α is the absorption coefficient, A is a proportionality constant, and E_g is the optical bandgap energy in electron volts (eV). As reported previously [19], the exponent n takes a value of 2 for allowed direct electronic transitions and 1/2 for indirect transitions. The optical bandgap energies of MgO, CeO₂, and MgO-CeO₂ NCs were determined using the Tauc relation by plotting $(\alpha h\nu)^2$ as a function of photon energy $h\nu$, assuming allowed direct transitions. Bandgap values were obtained by extrapolating the linear portion of the absorption edge to the energy axis. In all cases, the selected fitting regions showed good linearity near the absorption onset (Fig. S4), indicating reliable determination of the optical bandgaps. The clear linear behaviour of the Tauc plots further validates the use of the direct transition model for MgO, CeO₂, and MgO-CeO₂ NCs [25,32].

3.3. FT-IR analysis

FT-IR spectroscopy provides insight into the types of chemical bonds and functional moieties present in the synthesized catalyst. It is particularly useful for identifying functional groups that facilitate adsorption of specific contaminants onto the surface of MgO-CeO₂ NCs. Fig. 1d shows the FT-IR spectrum of MgO-CeO₂ NCs synthesized using *Sa* leaf extract, with vibrational frequencies recorded in the 4000–500 cm⁻¹ range. Several characteristic peaks of the *Sa* leaf extract were observed at 3328.89, 2969.88, 2889.17, 1658.41, 1373.21, 1094.93, 1045.73, 878.14, 804.35, and 606.01 cm⁻¹. The strong absorption peak at 3328.89 cm⁻¹ corresponds to O–H stretching vibrations of alcohol and phenolic groups [28]. Peaks around 2969.88 and 2889.17 cm⁻¹ are attributed to symmetric and asymmetric C–H stretching vibrations of CH₂/CH₃ groups. The band near 1658.41 cm⁻¹ arises from asymmetric C–H bending in methyl groups and aliphatic C–C stretching. Signals at

1094.93 and 1045.73 cm^{-1} are due to C—O stretching vibrations of ester functionalities, while the peak at 1373.21 cm^{-1} corresponds to asymmetric bending of C—H bonds. Aromatic ring vibrations appear as bands between 878.14 and 804.35 cm^{-1} , and the peak at 606.01 cm^{-1} is assigned to out-of-plane C—H bending in benzene rings [11,19].

Metal-oxygen (M—O) vibrations are observed in the lower wavenumber region. The signal at 557.31 cm^{-1} corresponds to Ce—O stretching, while Mg—O vibrations appear at 682.38 cm^{-1} . The overlap of these metal-oxide peaks in the 530–950 cm^{-1} region confirms the coexistence of MgO and CeO₂ in the composite [11,33]. Additionally, observed shifts and reduced intensities of certain absorption bands indicate interactions between the metal oxides and phytochemicals from the plant extract, further supporting the successful formation of MgO-CeO₂ NCs.

3.4. XPS analysis

HR-XPS was employed to analyze the surface oxidation states, chemical composition, and electronic configurations of elements in the MgO-CeO₂ NCs (Fig. 1). The wide-scan HR-XPS spectrum (Fig. 1e) provides an overview of the elemental composition of the composite. The deconvoluted Mg 2s spectrum (Fig. 1f) shows two distinct peaks at binding energies of 89.93 eV and 88.29 eV, corresponding to MgO and Mg(OH)₂, respectively. This indicates that magnesium predominantly exists in the Mg²⁺ oxidation state within the MgO lattice, with a minor fraction present as Mg(OH)₂ [11,25].

The Ce 3d spectrum (Fig. 1g) was deconvoluted into two pairs of peaks, each separated by 18.7 eV, confirming the presence of mixed valence states. The binding energy range for Ce 3d_{5/2} is 881.4–899.8 eV, while that for Ce 3d_{3/2} is 906.5–915.6 eV. Peaks at 881.4, 887.7, 897.4, 899.8, 906.5, and 915.6 eV correspond to the spin-orbit components of Ce³⁺ and Ce⁴⁺ ions, whereas satellite peaks at 883.7 and 901.8 eV confirm the coexistence of both oxidation states. The presence of Ce³⁺ and Ce⁴⁺ ions indicates oxygen vacancies within the crystal lattice, and binding energy shifts suggest charge transfer interactions between MgO and CeO₂. This mixed valence enhances the redox potential and surface reactivity of the composite, contributing to its superior photocatalytic performance [19,34,35].

In the O 1s spectrum (Fig. 1h), two main peaks were observed at ~529.8 eV and 531.2 eV, corresponding to lattice oxygen (O²⁻) associated with Ce³⁺/Ce⁴⁺ and Mg²⁺, and to surface hydroxyl groups and oxygen vacancies, respectively. These findings confirm the presence of structural oxygen, surface-bound oxygen species, and oxygen-deficient sites, which play a crucial role in enhancing the material's oxygen storage capacity and photocatalytic efficiency [35,36].

3.5. FE-SEM and HR-TEM analysis

The surface topography and microstructure of MgO-CeO₂ NCs were examined using FE-SEM (Fig. 2a). At a magnification of 100,000 $\times$, the morphology revealed the formation of crystallites with detailed surface texture. The NCs predominantly exhibited a spherical morphology, consisting of small, uniformly distributed particles that appeared aggregated and adhered to one another [33]. Such aggregation likely results from the densification process inherent to plant-mediated synthesis of MgO-CeO₂ NCs [35]. Despite this, only a moderate degree of clustering was observed. The well-defined structural features suggest good permeability, while the surface roughness indicates even dispersion of the NCs across the substrate [34].

HR-TEM images revealed multiple coexisting morphologies, including nanorods, hexagonal structures, and spherical particles, reflecting the distinct shapes of both MgO and CeO₂ within the composite (Figs. 2b–c) [33]. Particle size analysis using ImageJ software (Fig. 2d) estimated an average particle size of ~20.69 nm. SAED patterns (Fig. 2e) exhibited concentric diffraction rings, confirming the highly polycrystalline nature of the NCs. Crystal planes corresponding to

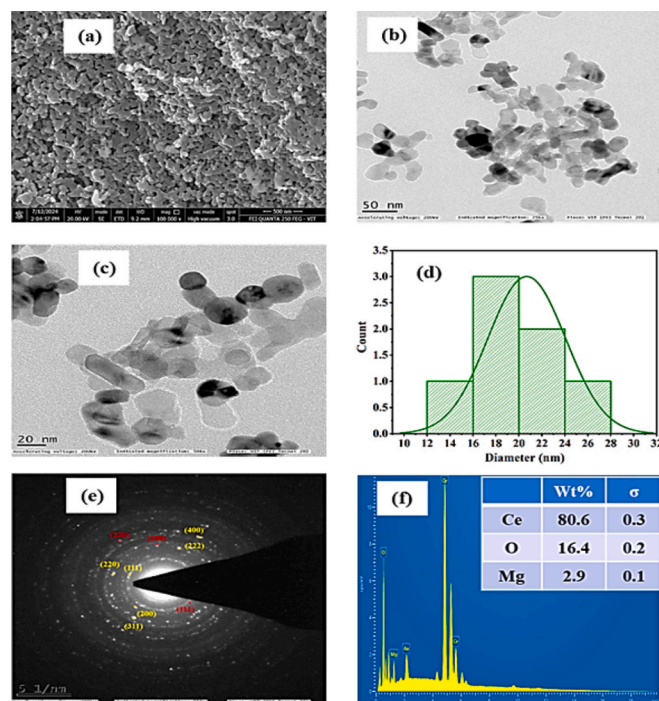


Fig. 2. (a) FE-SEM image, (b, c) HR-TEM images at distinct magnifications, (d) particle size distribution, (e) SAED pattern and (f) EDX spectra of MgO-CeO₂ NCs.

MgO are marked in red, while those of CeO₂ are denoted in yellow, clearly demonstrating the coexistence of two distinct crystalline phases [30]. HR-TEM images of individual MgO and CeO₂ NPs are provided in Figs. S5a–b.

EDX spectroscopy was employed to assess the elemental composition of the MgO-CeO₂ NCs (Fig. 2f). Distinct peaks corresponding to magnesium, cerium, and oxygen were observed, confirming the successful formation of the composite. The pronounced intensity of these peaks indicates the elements exist in their respective elemental states. Quantitative analysis revealed that the NCs contained approximately 2.9%, 80.6%, and 16.4% by weight of magnesium, cerium, and oxygen, respectively [30].

3.6. BET and TGA analysis

The surface area and porosity of MgO-CeO₂ NCs were evaluated using the BET method through N₂ adsorption-desorption experiments to determine the specific surface area (SSA), pore size, pore volume, and isotherm characteristics [37]. The N₂ adsorption isotherm profile of the NCs is shown in Fig. 3a. Surface area and porosity play a crucial role in heterogeneous catalysis, as they directly influence photocatalytic efficiency by enhancing charge-carrier transport and increasing the availability of surface reaction sites [38].

The results indicate that the MgO-CeO₂ NCs exhibit a type IV adsorption isotherm with an H3 hysteresis loop (Fig. 3a), characteristic of mesoporous materials. Type IV isotherms arise from monolayer-multilayer adsorption followed by capillary condensation within mesopores in the 2–50 nm range [36,38]. The BET surface area of the MgO-CeO₂ NCs is 87.03 m² g⁻¹, with an average pore diameter of 4.71 nm and a total pore volume of ~0.504 cc g⁻¹.

These values are significantly enhanced compared with individual MgO NPs, which show a lower surface area (40.773 m² g⁻¹) with a relatively large average pore width (29.013 nm) and pore volume (0.431 cc g⁻¹) [11], and are also higher than those of CeO₂ NPs, which exhibit a surface area of 77.033 m² g⁻¹, a smaller pore volume (0.150 cc g⁻¹), and an average pore diameter of 4.670 nm [19]. The improvement

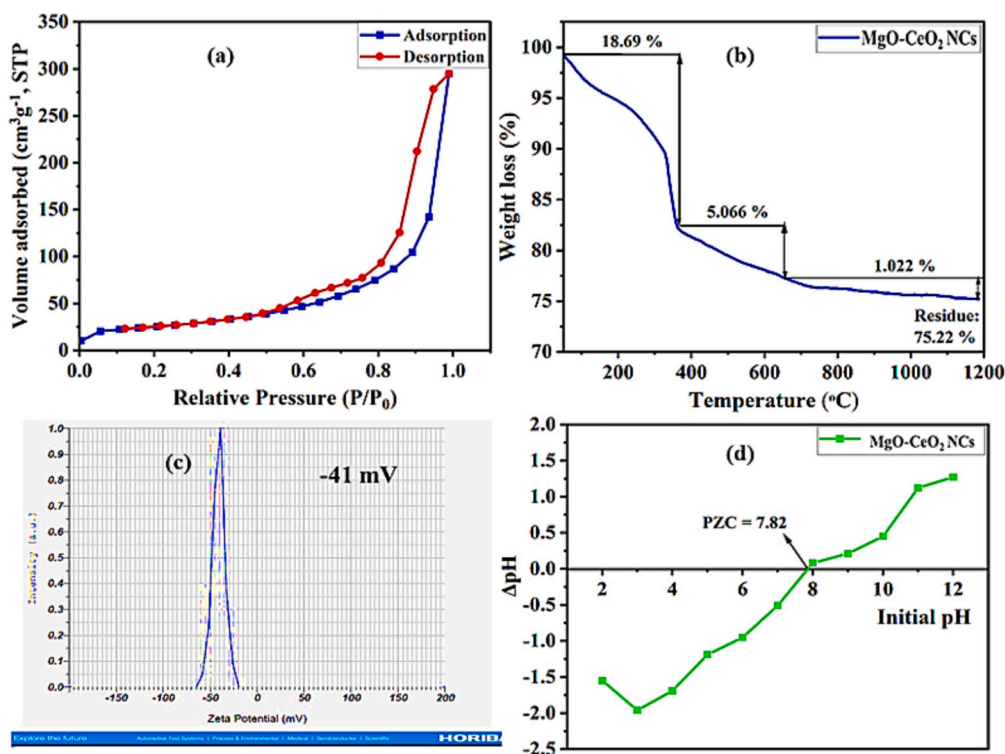


Fig. 3. (a) N₂ adsorption-desorption isotherm, (b) TGA analysis, (c) zeta potential and (d) pH_{PZC} analysis of MgO-CeO₂ NCs.

in surface area and pore characteristics upon composite formation can be attributed to strong interfacial interactions between the MgO and CeO₂ phases, which suppress particle agglomeration and promote the development of interconnected mesoporous networks. Importantly, the mesopore diameter of ~ 4.7 nm lies within an optimal range for photocatalytic applications, enabling efficient diffusion of dye molecules and reactive intermediates while maintaining a high density of accessible surface-active sites. In addition, the mesoporous architecture facilitates deeper light penetration and shortens the diffusion length of photo-generated charge carriers, thereby reducing electron-hole recombination. Consequently, the high surface area and optimized pore structure enhance adsorption capacity, improve reaction kinetics, and ultimately boost the photocatalytic performance of MgO-CeO₂ NCs [26,39].

The thermal stability of MgO-CeO₂ NCs was evaluated using TGA, as shown in Fig. 3b. Before calcination under N₂, the measurements were performed on air-dried samples at a heating rate of 20 °C/min. The TGA curve shows an initial weight loss of about 18.69 wt% between 100 °C and 350 °C, attributed to the evaporation of physically adsorbed moisture and the removal of surface hydroxyl groups [33,40]. A further mass loss of approximately 5.07 wt% occurs between 350 °C and 650 °C, mainly due to the decomposition of residual organic species originating from the Sa-leaf extract, including unreacted biomolecules and oxygen-containing functional groups bound to the surface [11,19]. Beyond 650 °C, the curve becomes nearly stable, with only a slight additional weight loss of around 1.30 wt%, demonstrating the remarkable thermal stability of the synthesized MgO-CeO₂ NCs at elevated temperatures.

3.7. Zetapotential and pH_{PZC} analysis of MgO-CeO₂ NCs

Zeta potential (ZP) measurements were carried out to evaluate the electrostatic interactions between the nanoparticles and the surrounding medium, which govern colloidal stability. The surface charge and dispersion behaviour of MgO-CeO₂ NCs calcined at 700 °C were analyzed using a zeta sizer. Nanomaterials exhibiting high absolute zeta potential values typically possess strong electrostatic stabilization, favouring long-term suspension stability [11].

In the present study, the MgO-CeO₂ NCs showed a ZP value of -41 mV (Fig. 3c), indicating excellent colloidal stability. The pronounced negative surface charge can be attributed to strong interfacial interactions between the MgO and CeO₂ phases and the presence of surface hydroxyl and defect sites, which enhance electrostatic repulsion among particles [25,41].

The point of zero charge (pH_{PZC}) is an important parameter for understanding the adsorption behaviour of nanomaterials, as it represents the pH at which the surface possesses no net electrical charge. At this condition, electrostatic attraction or repulsion is minimized, and adsorption mainly occurs through non-electrostatic interactions. When the solution pH is lower than the pH_{PZC}, the surface of MgO-CeO₂ NCs becomes positively charged, whereas at pH values higher than the pH_{PZC}, the surface turns negatively charged, which strongly affects interactions with charged dye molecules [33].

In this work, the pH drift method was used to determine the pH_{PZC} of MgO-CeO₂ NCs. A 0.1 M NaCl solution was adjusted to initial pH values between 2 and 12 using 0.1 N HCl or NaOH. Then, 0.1 g of MgO-CeO₂ NCs was added to each solution, while blank solutions were prepared without the catalyst. The suspensions were shaken at 150 rpm for 18 h, filtered, and the final pH (pH_F) was measured [41]. The pH_{PZC} was obtained from the plot of $\Delta\text{pH} = \text{pH}_i - \text{pH}_F$ versus pH_i, where the curve crosses $\Delta\text{pH} = 0$ (Fig. 3d). The MgO-CeO₂ NCs showed a pH_{PZC} of 7.82, indicating a positively charged surface at pH < 7.82 and a negatively charged surface at pH > 7.82 [26].

3.8. Photoluminescence analysis of MgO-CeO₂ NCs

Photoluminescence (PL) spectroscopy is an effective technique for investigating the recombination behaviour, mobility, and transfer efficiency of photogenerated charge carriers in semiconductor materials. PL emission originates from the radiative recombination of excited electrons from the conduction band (CB) to the valence band (VB). Consequently, the PL intensity serves as an indicator of the recombination rate of electron-hole pairs, where lower emission intensity reflects efficient charge separation and enhanced photocatalytic performance [26,42].

The PL emission spectra of MgO-CeO₂ NCs and the individual oxides (MgO and CeO₂) were recorded at an excitation wavelength around 450 nm under ambient conditions (Fig. S6). A comparative evaluation reveals that MgO and CeO₂ NPs exhibit higher PL intensities, whereas the MgO-CeO₂ NCs show a markedly quenched emission. This strong PL quenching confirms that the heterojunction formed between MgO and CeO₂ effectively suppresses the radiative recombination of photoinduced electrons and holes.

The reduced PL intensity of MgO-CeO₂ NCs is attributed to efficient interfacial charge transfer between the MgO and CeO₂ phases, which prolongs the lifetime of charge carriers and promotes their participation in surface redox reactions. As a result, the MgO-CeO₂ NCs demonstrate superior photocatalytic efficiency compared to the individual components. Overall, the PL results highlight the critical role of heterostructure formation in enhancing charge separation and photocatalytic activity.

3.9. Evaluation of photocatalytic performance of MgO-CeO₂ NCs

The photocatalytic performance of MgO-CeO₂ NCs was systematically evaluated for the visible-light-driven degradation of cationic RhB and anionic EY, monitored using UV-vis absorption spectroscopy. The characteristic absorption maxima of RhB (554 nm) and EY (515 nm) were recorded at regular time intervals under optimized conditions (Fig. 4). To investigate their influence on degradation efficiency, key operational parameters were varied, including solution pH (2–12), initial dye concentration (10–50 ppm), catalyst dosage (0.2–1.0 mg/mL), and irradiation source. The light sources comprised natural sunlight, visible-light tungsten lamps (500 W and 300 W) with measured intensities of 265 and 59 Lux, respectively, and a UV mercury lamp (250 W, 365 nm) delivering an intensity of 380 Lux at the reactor surface. Before irradiation, an adsorption-desorption equilibrium was established in the dark, where the decrease in absorbance was $\leq 4\%$ for both RhB and EY, indicating negligible dye adsorption on the catalyst surface (Figs. S7a,b). Control experiments carried out under light without the

photocatalyst showed $\leq 2\%$ dye removal, confirming that direct photolysis was insignificant. Under optimal conditions, a continuous decrease in absorbance was observed, leading to nearly complete degradation of RhB within 120 min (Fig. 4a) and EY within 80 min (Fig. 4b). In contrast, experiments performed in the absence of MgO-CeO₂ NCs exhibited minimal changes in dye concentration, underscoring the essential role of the photocatalyst in promoting degradation. Overall, these results demonstrate the strong potential of MgO-CeO₂ NCs for wastewater remediation, confirming their high photocatalytic efficiency under visible-light irradiation.

3.9.1. Impact of pH solution

The effect of solution pH (2–12) on the photocatalytic degradation of cationic RhB and anionic EY using MgO-CeO₂ NCs was systematically investigated. As shown in Fig. 4, the degradation efficiency of both dyes strongly depends on pH. During pH optimization (2, 4, 7, 9, and 12), all other parameters, including dye concentration (10 ppm), catalyst dosage (1 mg/mL), and visible-light source (500 W), were maintained constant at ambient temperature. Before irradiation, the dye solutions were stirred in the dark for 60 min (RhB) and 30 min (EY) to establish adsorption-desorption equilibrium. Subsequently, the suspensions were irradiated under visible light for 120 min (RhB) and 80 min (EY), and degradation was monitored by measuring the absorbance at 554 nm for RhB and 515 nm for EY, respectively. The corresponding first-order kinetic plots are presented in Fig. 4c and d. For RhB, the degradation efficiency increased with increasing pH, reaching a maximum value of 94.69% at pH 12. This behaviour is governed by the point of zero charge (pH_{PZC}) of MgO-CeO₂ NCs, determined to be 7.82. At pH values above the pH_{PZC}, the catalyst surface becomes negatively charged, which enhances electrostatic attraction toward the positively charged RhB molecules. This promotes greater adsorption of RhB on the catalyst surface and facilitates more effective interaction with photogenerated reactive species, leading to improved degradation. In contrast, under acidic conditions (pH 2), the MgO-CeO₂ surface is positively charged and

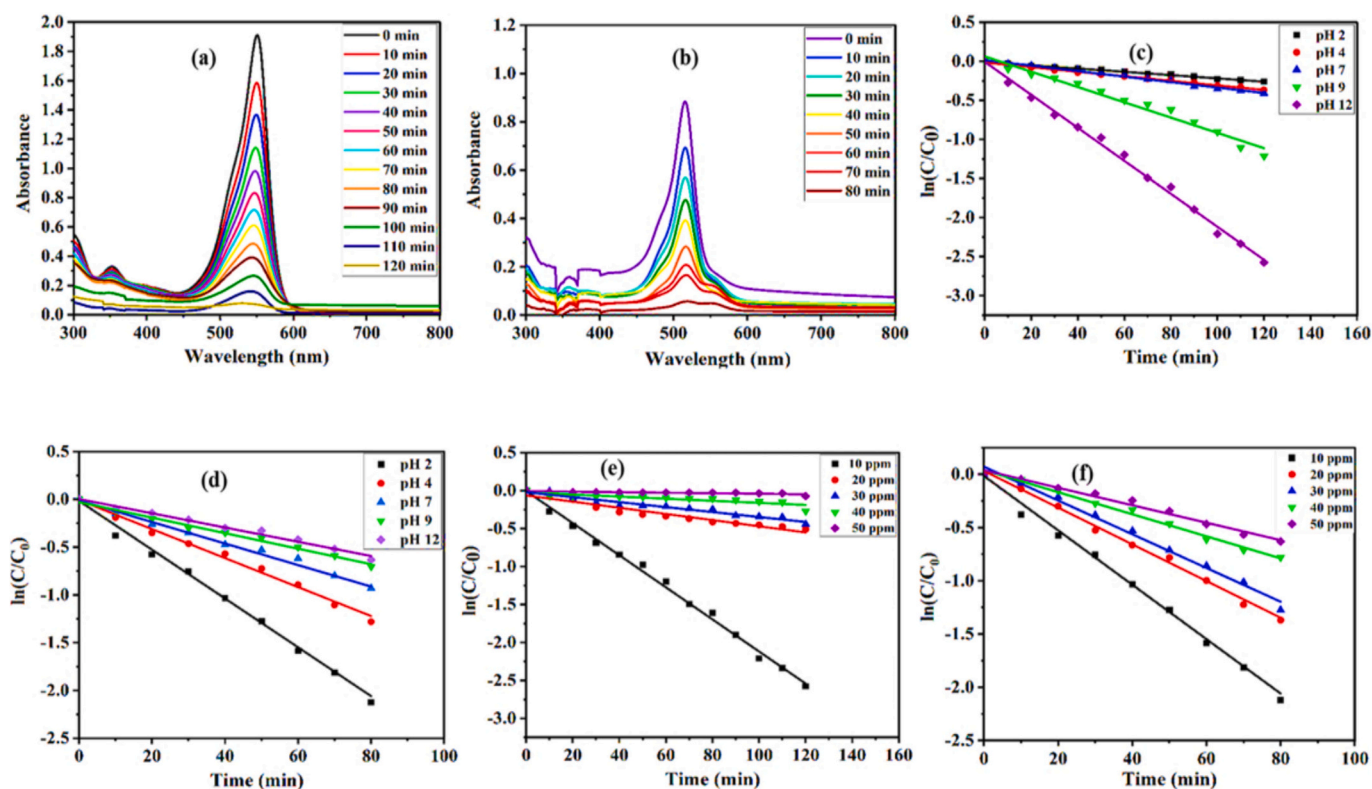


Fig. 4. UV-vis absorption spectra for (a) RhB, (b) EY dyes degradation, Kinetic plots for the degradation of (c) RhB, and (d) EY dyes at different pH values, Kinetic plots for degradation of (e) RhB, and (f) EY dyes at different concentrations of dyes.

electrostatically repels cationic RhB, thereby reducing adsorption and lowering the degradation efficiency to 25.79% [11,19,43].

For EY, an anionic dye, the highest degradation efficiency (91.02%) was obtained at pH 2, where the catalyst surface is positively charged and strongly attracts negatively charged EY molecules. As the pH increased from 2 to 12, the surface charge of MgO-CeO₂ NCs became increasingly negative, resulting in electrostatic repulsion with EY and a gradual decrease in degradation efficiency to 55.58% [25,41,44]. These results clearly indicate that pH_{PZC} plays a decisive role in controlling the surface charge of the photocatalyst and, consequently, the electrostatic interactions between the catalyst and dye molecules. The overall pH-dependent degradation trends for both dyes are summarized in Table S1.

At neutral pH (pH 7), the MgO-CeO₂ NCs still exhibit appreciable photocatalytic activity toward both RhB and EY (Fig. 4c and d). The degradation efficiencies reach 33.55% for RhB and 58.24% for EY (Tables 2 and 3). The relatively lower efficiencies near neutral pH are attributed to weaker electrostatic interactions between the dye molecules and the catalyst surface close to the pH_{PZC}, which limits adsorption and reduces the availability of reactive radicals [41]. Nevertheless, the noticeable degradation at pH 7 confirms that MgO-CeO₂ NCs remain catalytically active under environmentally relevant wastewater conditions.

3.9.2. Impact of RhB and EY concentrations

Photocatalytic tests occurred at pH 12 (RhB) and pH 2 (EY) using MgO-CeO₂ NCs (1 mg/mL) with increasing dye concentrations of 10, 20, 30, 40 and 50 ppm. The degradation efficiency decreased significantly as the dye content rose. As shown in Fig. 4e, the degradation efficiency of RhB decreased markedly from 94.69% at an initial concentration of 10 ppm to 7.01% at 50 ppm within a 120 min irradiation period, following first-order kinetic behaviour. Similarly, EY exhibited a maximum degradation efficiency of 91.02% at 10 ppm, which progressively declined to 54.17% as the dye concentration increased to 50 ppm within a 90 min time interval (Fig. 4f). This decline in degradation rate with larger dye concentrations might be due to the catalyst surface becoming saturated contains too many dye molecules, limiting the availability of

Table 2

Effect of reaction conditions on the photocatalytic degradation efficiency of RhB using MgO-CeO₂ NCs.

RhB dye	Parameters	Time (min)	Degradation efficiency (%)	Rate constants (k) = ($\times 10^{-4}$ s ⁻¹)	R ² value
pH	2	120	25.79	0.80	0.9935
	4	120	30.81	1.11	0.9949
	7	120	33.55	1.34	0.9898
	9	120	79.31	3.75	0.9723
	12	120	94.69	8.10	0.9954
Concentration (ppm)	10	120	94.69	8.10	0.9954
	20	120	38.98	1.55	0.9498
	30	120	35.60	1.25	0.9423
	40	120	23.65	0.52	0.7418
	50	120	7.01	0.13	0.6502
Catalyst dose (mg/ mL)	0.2	120	23.99	0.80	0.9215
	0.4	120	29.18	1.04	0.9713
	0.6	120	29.34	1.48	0.9809
	0.8	120	73.20	3.80	0.9883
	1.0	120	94.69	8.10	0.9954
Light sources	250 W 365 nm UV light	70	98.08	18.94	0.9939
	500 W Visible light	120	94.69	8.10	0.9954
	300 W Visible light	120	71.47	1.96	0.9909
	Sunlight	120	30.60	1.10	0.9696

Table 3

Effect of reaction conditions on the photocatalytic degradation efficiency of EY using MgO-CeO₂ NCs.

EY dye	Parameters	Time (min)	Degradation efficiency (%)	Rate constants (k) = ($\times 10^{-4}$ s ⁻¹)	R ² value
pH	2	80	91.02	9.80	0.9976
	4	80	62.90	5.82	0.9841
	7	80	58.24	4.32	0.9835
	9	80	55.59	3.11	0.9888
	12	80	55.58	2.85	0.9860
Concentration (ppm)	10	80	91.02	9.80	0.9976
	20	80	74.59	0.66	0.9843
	30	80	72.02	0.60	0.9913
	40	80	61.78	0.39	0.9904
	50	80	54.17	0.31	0.9851
Catalyst dose (mg/ mL)	0.2	80	65.35	5.30	0.9910
	0.4	80	71.23	5.95	0.9957
	0.6	80	73.54	6.71	0.9945
	0.8	80	91.70	11.34	0.9967
	1.0	80	91.02	9.80	0.9976
Light sources	250 W 365 nm UV light	70	94.42	13.62	0.9969
	500 W Visible light	80	91.70	11.34	0.9976
	300 W Visible light	80	44.43	2.85	0.9959
	Sunlight	80	32.61	1.78	0.9772

active sites for photocatalytic processes. Furthermore, high dye concentrations make the solution more turbid, lowering light penetration and, as a result, restricting the interaction between catalyst surface and incoming photons. As a result, photocatalytic activity decreases due to decreased production of reactive species [40,41]. As a result, an optimum dye concentration of 10 ppm for both RhB and EY was discovered to be suitable for effective photocatalytic degradation under the specified conditions [11].

3.9.3. Impact of photocatalyst loading

The amount of catalyst strongly influences photocatalytic degradation efficiency, making the optimization of catalyst dosage essential to achieve maximum performance without unnecessary excess. In this study, different loadings of MgO-CeO₂ NCs (0.2–1.0 mg/mL) were evaluated for the degradation of RhB and EY at their respective optimal pH values (pH 12 for RhB and pH 2 for EY) using a fixed dye concentration of 10 ppm. The reactions were conducted for 120 min for RhB and 80 min for EY. As shown in Fig. 5a and b, photocatalytic activity increased with higher catalyst loading for both dyes. For RhB, increasing the catalyst from 0.2 to 1.0 mg/mL enhanced degradation efficiency from 23.99% to 94.69% within 120 min. Similarly, EY degradation improved markedly, reaching a maximum efficiency of 94.42% at 0.8 mg/mL within 80 min. This improvement is attributed to the increased availability of surface-active sites, which promotes the generation of reactive oxygen species ($\bullet$ OH and $\bullet$ O₂⁻) and accelerates dye decomposition.

Beyond the optimal dosage, photocatalytic efficiency declined. Excess catalyst increases solution turbidity and light scattering, reducing photon penetration into the reaction medium. Additionally, higher loadings may cause particle aggregation, lowering the effective surface area and suppressing catalytic activity [11,19,25,41]. These light-shielding and interparticle effects are well known to limit photocatalytic performance at elevated catalyst concentrations. Accordingly, the optimal MgO-CeO₂ NCs dosage was determined to be 1.0 mg/mL for RhB and 0.8 mg/mL for EY, achieving maximum degradation efficiency while maintaining favourable reaction kinetics.

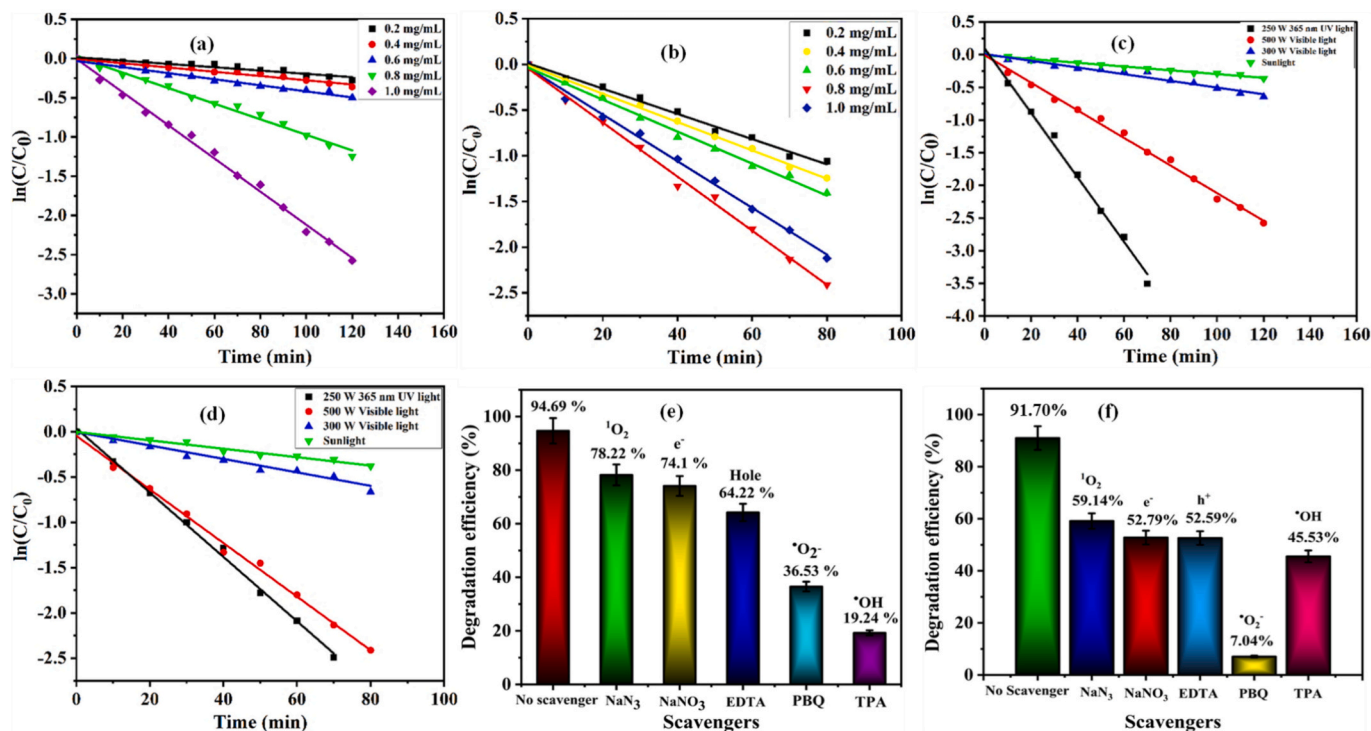


Fig. 5. Kinetic plots for the degradation of (a) RhB, and (b) EY dyes at different catalyst dosages, Kinetic plots for the degradation of (c) RhB, and (d) EY dyes at different light sources, Efficacy of several scavengers for (e) RhB, and (f) EY dyes.

3.9.4. Impact of light intensity

Light intensity is a key factor governing the photocatalytic degradation rate of dyes. Fig. 5c and d present the first-order kinetic plots for RhB and EY degradation under different irradiation sources, including UV light (250 W, 365 nm; 380 Lux), visible light from tungsten lamps (300 W and 500 W, 400–800 nm; 59 Lux and 265 Lux, respectively), and natural sunlight. During these experiments, the dye concentration, catalyst dosage, and pH were maintained at their optimized values. An increase in degradation rate was observed with increasing light intensity, which can be attributed to the enhanced generation of photo-induced charge carriers that promote the formation of reactive oxygen species responsible for dye decomposition. In contrast, at lower light intensities, the reduced photon flux limits charge-carrier excitation and increases electron-hole recombination, thereby lowering photocatalytic efficiency [45].

Among the tested sources, 500 W visible light showed an excellent photocatalytic performance, achieving degradation efficiencies of 94.69% for RhB after 120 min and 91.02% for EY after 80 min. UV irradiation exhibited faster kinetics, reaching 98.08% RhB degradation and 94.42% EY degradation within 70 min. The strong activity under visible light is mainly due to the reduced band gap of MgO-CeO₂ NCs, which enhances visible-light absorption and facilitates efficient excitation and separation of charge carriers [25,43]. The corresponding time-dependent degradation efficiencies, apparent rate constants, and correlation coefficients (R^2) under different irradiation conditions are summarized in Tables 2 and 3 for RhB and EY, respectively [46]. The photocatalytic performance of MgO-CeO₂ NCs with different weight ratios and the optimized 1:1 composite calcined at various temperatures is presented in Figs. S8a and S8b and Table S2. In addition, Table S3 compares the performance of reported MgO-CeO₂-based photocatalysts for organic dye degradation under light irradiation.

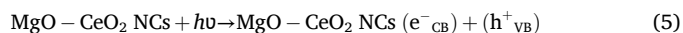
3.10. Radical scavenger experiment

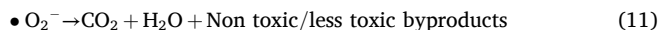
The photocatalytic efficiency of MgO-CeO₂ NCs depends on the effective interaction between dye molecules and the catalyst surface, as

well as the rapid generation of charge carriers upon light exposure [42]. To identify the ROS responsible for dye degradation, radical scavenger experiments were performed under visible-light irradiation. Specific scavengers were employed to selectively quench targeted ROS: EDTA for photogenerated holes (h^+), NaN₃ for singlet oxygen (1O_2), NaNO₃ for electrons (e^-), *p*-benzoquinone (PBQ) for superoxide radicals ($\bullet O_2^-$), and terephthalic acid (TPA) for hydroxyl radicals ($\bullet OH$) [11,19].

Dye solutions (RhB or EY) containing the optimized catalyst dosage were treated with each scavenger individually, maintaining a 1:1 ratio of scavenger to catalyst under standard photocatalytic conditions. The catalyst concentration was fixed at 1 mg/mL, and 1 mg of each scavenger was added separately, while the dye solution volume and irradiation conditions remained constant. Degradation efficiency was monitored using UV-vis spectrophotometry. Significant decreases in dye degradation in the presence of specific scavengers indicate the predominant ROS involved. For example, suppression of degradation with TPA and PBQ confirmed the key roles of $\bullet OH$ and $\bullet O_2^-$ radicals in the photocatalytic decomposition of RhB and EY, respectively (Fig. 5e and f).

Upon visible-light absorption, MgO-CeO₂ NCs capture photons ($h\nu$), exciting electrons (e^-) from the VB to the CB, generating electron-hole pairs. Due to the Type-I heterojunction between MgO and CeO₂, photogenerated electrons and holes from the CB and VB of MgO migrate to the CB and VB of CeO₂ [47]. The h^+ oxidize surface hydroxyl groups ($-OH$) or water molecules to produce highly reactive $\bullet OH$ radicals, while the CB e^- reduce dissolved O₂ to form $\bullet O_2^-$ radicals. These ROS synergistically attack the dye molecules, ultimately mineralizing them into CO₂, H₂O, and non-toxic by-products. This mechanism of ROS generation and electron-hole transfer is illustrated in Eqs. 5–11 [11,25,36].





To better understand the photocatalytic degradation mechanism of RhB and EY dyes over MgO-CeO₂ NCs, the CB and VB edge potentials were estimated using the Mulliken electronegativity approach, as expressed in Eqs. (12) and (13) [43]:

$$E_{CB} = X - E^e - 0.5E_g \quad (12)$$

$$E_{VB} = E_{CB} + E_g \quad (13)$$

where E_{CB} and E_{VB} represent the CB and VB potentials, respectively; E_g is the bandgap energy; X is the absolute electronegativity of the nano-composite; and E^e is the energy of free electrons on the hydrogen scale (4.5 eV), corresponding to the potential of electrons relative to the normal hydrogen electrode (NHE) [48,49]. Based on these calculated band positions, the proposed photocatalytic pathways for RhB and EY degradation over MgO-CeO₂ NCs are illustrated in Fig. 6a and b.

3.11. Recyclability and stability of MgO-CeO₂ NCs

To evaluate the reusability of MgO-CeO₂ NCs, the photocatalyst was recovered from the reaction mixture by centrifugation after each degradation experiment, thoroughly rinsed with DDW, and then dried at 80 °C for 12 h before reuse in subsequent photocatalytic cycles (Fig. 6c and d). The reusability tests show a gradual decrease in degradation efficiency with increasing reuse cycles. For RhB, the degradation efficiency slightly decreases from 94% in the first cycle to 87% in the fourth cycle, whereas for EY it decreases from 91% to 86% over the same cycles. This marginal loss in activity can be attributed to minor catalyst loss during the recovery and washing processes, blockage of surface-

active sites by adsorbed reaction intermediates, and partial aggregation of particles after repeated light irradiation [41]. To examine the structural integrity of the NCs, XRD analyses were performed on both fresh and reused catalysts after four photocatalytic cycles (Fig. 6e and f). The diffraction patterns reveal no noticeable changes in crystal structure or the appearance of new peaks, confirming the excellent structural stability and durability of the MgO-CeO₂ NCs. These results demonstrate that the MgO-CeO₂ NCs maintain high photocatalytic efficiency over multiple reuse cycles, indicating good recyclability and operational stability for the degradation of both RhB and EY dyes [11,25,43].

3.12. LC-MS profiling of decomposition products

The degradation by-products of RhB and EY dyes were analyzed by LC-MS using various solvent gradients composed of H₂O, EtOH, ACN, and IPA as mobile phases, along with an unsprayed ionization source. The mass spectra revealed the molecular ions of RhB and EY, their characteristic fragmentation patterns, and the photocatalytic intermediates formed during degradation, as shown in Fig. 7a and b. The progressive breakdown of RhB and EY under visible-light irradiation in the presence of MgO-CeO₂ NCs is further corroborated by LC-MS results, which demonstrate the formation of numerous low-molecular-weight compounds (Fig. 7c and d) [11,24].

These degradation products are mainly generated through oxidative reactions involving $\bullet OH$ and $\bullet O_2^-$ radicals produced during the photocatalytic process. The proposed intermediate species formed along the degradation pathways are summarized in Table 4 and Table 5, indicating that RhB and EY are gradually mineralized into less hazardous compounds. Based on the experimentally observed intermediates, plausible degradation pathways were constructed, and additional intermediates were inferred from sequential bond cleavage and oxidation steps reported in the literature. Therefore, the molecules presented in the degradation mechanisms include both experimentally detected and theoretically proposed species.

Furthermore, the potential environmental impact of the parent dyes and the LC-MS identified intermediates were evaluated using ECOSAR software [25,41,43]. The predicted acute and chronic toxicity values

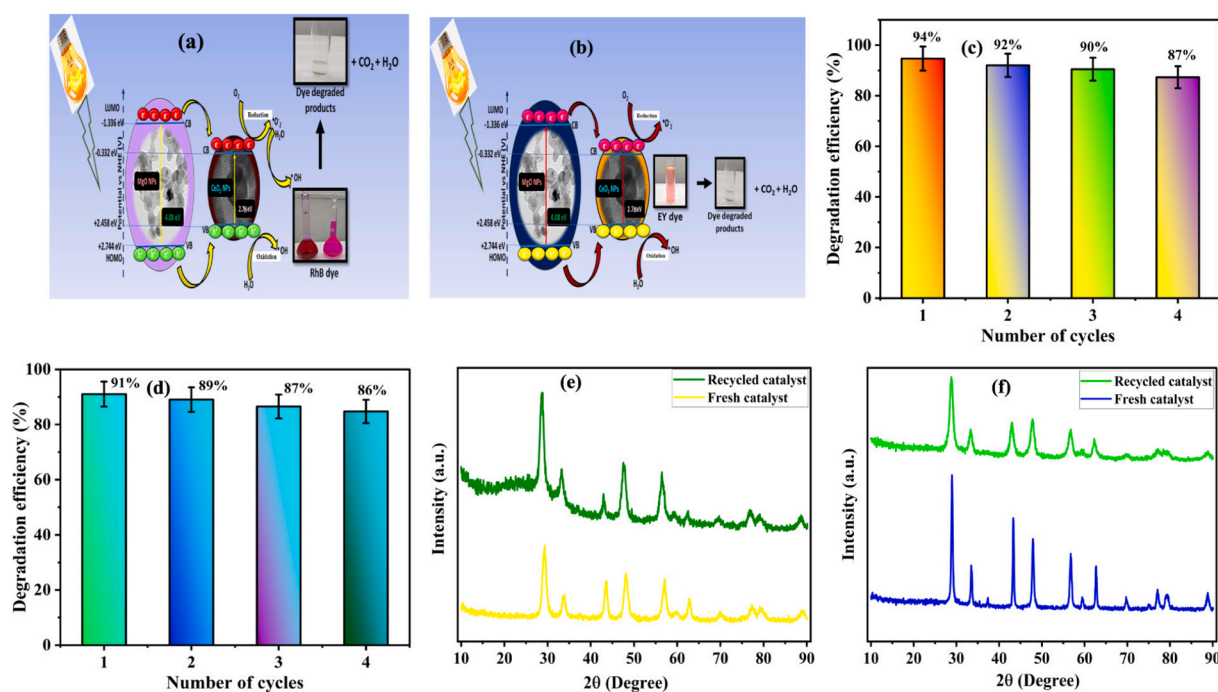


Fig. 6. Possible mechanism of photocatalytic degradation for (a) RhB, and (b) EY dyes by MgO-CeO₂ NCs, Recyclability studies of (c) RhB, and (d) EY dye degradation using MgO-CeO₂ NCs, XRD analysis of MgO-CeO₂ NCs, before and after breakdown of (e) RhB, and (f) EY dyes.

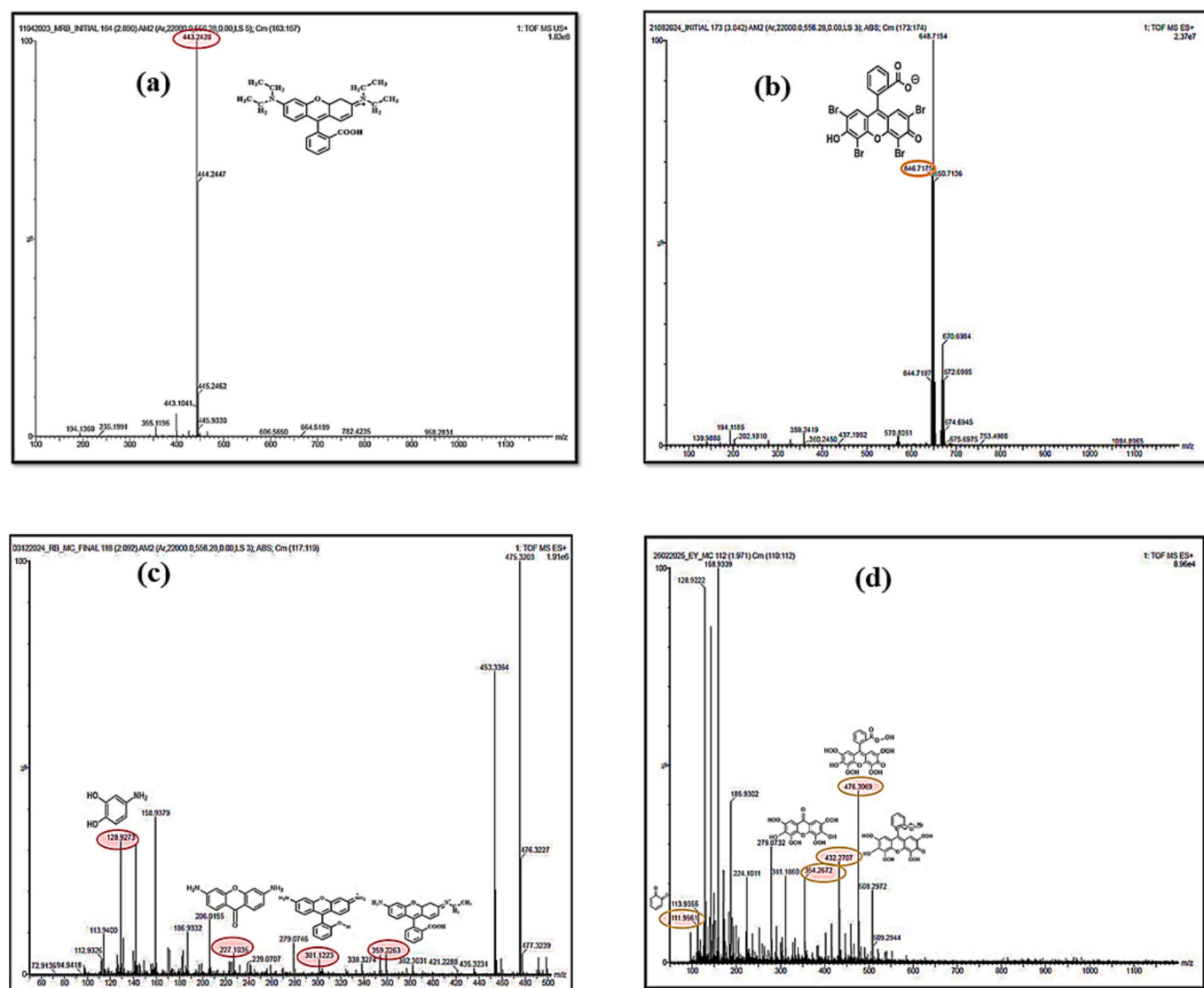


Fig. 7. LC-MS analysis of (a) RhB and (b) EY dyes before photocatalytic breakdown, LC-MS analysis of (c) RhB and (d) EY dyes after photocatalytic degradation.

(Tables S3 and S4) suggest that the major degradation products exhibit significantly lower toxicity toward aquatic organisms compared with RhB and EY, confirming that the photocatalytic process leads to effective detoxification. The corresponding theoretical degradation mechanisms for RhB and EY are depicted in Figs. S9 and S10.

4. Conclusions

In this study, MgO-CeO₂ NCs were successfully synthesized, and the effect of calcination temperature on their band gap was systematically investigated. The NCs were comprehensively characterized by XRD, FT-IR, XPS, FE-SEM with EDX, HR-TEM, UV-vis spectroscopy, BET, TGA analysis, and zeta potential measurements. HR-TEM analysis revealed an average particle size of around 20.69 nm, while XRD confirmed the formation of a *fcc* crystalline structure for both MgO and CeO₂ components. The optical band gap was determined using Tauc plots. The photocatalytic performance of MgO-CeO₂ NCs was evaluated through the degradation of cationic RhB and anionic EY dyes under various conditions, including solution pH, dye concentration, catalyst dosage, and light source. Optimal degradation was achieved at pH 12 for RhB and pH 2 for EY, with catalyst dosages of 1 mg/mL and 0.8 mg/mL, respectively. Radical scavenger experiments identified •OH and •O₂⁻ as the primary reactive species driving dye degradation. LC-MS analysis

confirmed the formation of low-molecular-weight intermediates, indicating stepwise mineralization of the dye molecules. Furthermore, recyclability studies demonstrated that the MgO-CeO₂ NCs retained excellent photocatalytic performance over multiple cycles, highlighting their stability and effectiveness. Overall, these results demonstrate that MgO-CeO₂ NCs are durable and efficient photocatalysts with significant potential for wastewater treatment applications.

5. Limitations and future outlook

This study was conducted using model dye solutions, and the influence of inorganic ions and natural organic matter in real wastewater was not evaluated. Long-term continuous operation and large-scale synthesis were also not explored. Future work will focus on testing MgO-CeO₂ NCs in real wastewater using continuous-flow photoreactors and optimizing the synthesis for scalable production, aiming toward practical wastewater treatment applications [19,25,50].

CRediT authorship contribution statement

Nethra Kuruthukulangara: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **D. Thirumalai:** Validation, Data curation. **Umesh Narasimha Murthy:** Formal analysis.

Table 4

Photocatalytic degradation intermediates of RhB identified by LC-MS analysis.

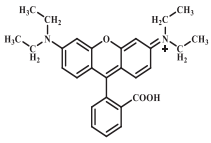
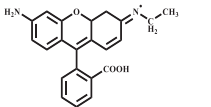
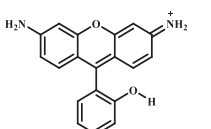
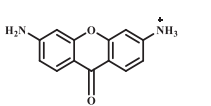
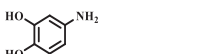
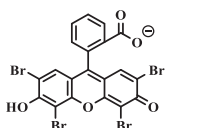
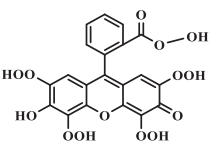
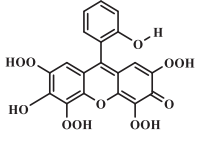
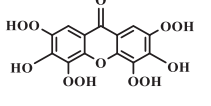
Sl. No.	Product structure	LC-MS mass values (m/z)	Theoretical mechanism mass values (m/z)	Molecular form
1		m/z 443	m/z 443	$C_{28}H_{31}N_2O_3^+$
2		m/z 359	m/z 360	$C_{22}H_{20}N_2O_3^+$
3		m/z 301	m/z 303	$C_{19}H_{15}N_2O_2^+$
4		m/z 227	m/z 227	$C_{13}H_{11}N_2O_2^+$
5		m/z 128	m/z 124	$C_6H_7NO_2$

Table 5

Photocatalytic degradation intermediates of EY identified by LC-MS analysis.

Sl. No.	Product structure	LC-MS mass values (m/z)	Theoretical mechanism mass values (m/z)	Molecular form
1		m/z 646	m/z 646	$C_{20}H_{12}Br_4O_5^-$
2		m/z 476	m/z 476	$C_{20}H_{12}O_{14}$
3		m/z 432	m/z 432	$C_{19}H_{12}O_{12}$
4		m/z 354	m/z 356	$C_{13}H_8O_{12}$
5		m/z 111	m/z 108	$C_6H_4O_2$

Werayut Srituravanich: Validation. **I.V. Asharani:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jwpe.2026.109790>.

Data availability

Data will be made available on request.

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